

Dr. Thejus Mahajan
Reventlowstraße 17
22605 Hamburg
email – thejusmahajanmarrat@gmail.com

03.04.2021

Prof. Dr. Inga Hense
Palmaille 9
22767 Hamburg

Dear Prof. Dr. Inga Hense,

It is with great excitement and interest that I am writing this mail to join your group in the development of numerical marine ecosystem model making use of the powerful “Resurrection paradigm” for the project titled “Modelling the evolutionary response of cyanobacteria to global warming”.

I have completed my Ph.D. in astrochemistry under the guidance of Dr. Karine Béroff, Institut des Sciences Moléculaires d'Orsay, Orsay France. My thesis is titled “Excitation and fragmentation of C_nN^+ ($n = 1-3$) molecules in collisions with He atoms at intermediate velocity; fundamental aspects and application to astrochemistry” in December 2018.

My principal motivation is my interest in the history and future of sustained evolution of ecosystem under variable conditions, especially under the so called “climate change”. For me, the position would mean a much-awaited career transition in to evolutionary biology from astrochemistry. For the team, I showcase my experience in numerical modelling during my work at M.Sc. (Physics) and Ph.D. (Astrochemistry), my willingness to learn and assimilate to the interdisciplinary field of evolutionary biology and team spirit to play my part to work on a common goal.

Working on the theoretical modelling of collision for my Ph.D. has given me the opportunity to develop strength and skills that are transferable and relevant to your project. For my Ph.D., I played a leading role in running the experiments, analysing terra-bytes of data from the silicon detectors, and developing a theoretical model for the collision based of CTMC (Classical Trajectory Monte Carlo) method and IAE (Independent Atom Electron) model. I successfully did a numerical simulation of fragmentation of molecules after collision with Fortran to find the loss rate of fragments due to an incorrectly positioned detector (human error).

This project is an excellent opportunity to learn and acquire new skills for my transition to evolutionary biology by contributing to the development and improvement of modelling codes to study the evolutionary response of cyanobacteria in aquatic ecosystem with a new set of dynamic factors such as global warming. My work during

the Ph.D. also contributed to the theoretical method of collision process apart from the experiments and I strive to contribute on a similar scale to this project.

Kindly consider my application for this project. Thank you for your time and I look forward to hearing back from you.

I attach my CV along with this cover letter. I am also available in speaking with you for a personal interview. I could also visit your lab at your convenience since I am staying very close to DESY with my wife who is working there as a post-doc.

Sincerely,
Dr. Thejus Mahajan

Dr. Thejus Mahajan

CONTACT Email : thejusmahajanmarrat@gmail.com
INFORMATION Ph: +91-7994471440
 Reventlowstraße 17
 22605 Hamburg, Germany

DATE OF BIRTH 30/September/1992

CAREER OBJECTIVE To apply myself fully on a goal oriented task through methodical approach and to realize mastery over method and means by continuous learning.

CAREER INTERESTS Numerical modelling of ion-atom collision systems, Application of physics to biology, Astrobiology - Origin and evolution of building blocks of life (suspect - molecules in ISM), Data analysis of collision experiments, Atom - ion interaction, ion-atom collision experiments, Classical Trajectory Monte Carlo Method.

- PUBLICATIONS
1. T Mahajan, K Béroff, B Pons, C Illescas, M Chabot, T Idbarkach, A Jorge, N F Aguirre, and S Diaz-Tendero. Energy deposit by electron excitation in C_nN projectiles ($n=1-3$) colliding at intermediate velocity with He atoms : semi-empirical estimates and calculations. *Journal of Physics: Conference Series*, 1412:142026, jan 2020
 2. T IdBarkach, M Chabot, K Béroff, S Della Negra, J Lesrel, F Geslin, A Le Padellec, T Mahajan, and S Díaz-Tendero. Breakdown curves of CH_2 (+), CH_3 (+), and CH_4 (+) molecules. construction and application to electron collisions and UV photodissociation. *Astronomy & Astrophysics*, 628:A75, 2019
 3. Thejus Mahajan, Karine Beroff, Bernard Pons, Clara Illescas, Marin Chabot, Tijani IdBarkach, Thibaut Launoy, Arnaud Le Padellec, Aurelie Jallat, Alba Jorge, Nestor Aguirre, and Sergio Diaz-Tendero. Excitation, ionization, neutralization and anionic production in collisions of C^+ , N^+ and C_nN^+ ($n=1-3$) with he atoms at 2.2 a.u velocity; cross sections and dissociation branching ratios. *Journal of Physics B: Atomic, Molecular and Optical Physics*, 2019
 4. Tijani IdBarkach, Thejus Mahajan, Marin Chabot, Karine Béroff, Néstor F. Aguirre, Sergio Diaz-Tendero, Thibaut Launoy, Arnaud Le Padellec, Luc Perrot, and Ma' Semiempirical breakdown curves of $C_2N^{(+)}$ and $C_3N^{(+)}$ molecules; application to products branching ratios predictions of physical and chemical processes involving these adducts. *Molecular Astrophysics*, 12:25 – 32, 2018
 5. T. Mahajan, T. Id Barkach, N.F. Aguirre, M. Alcamí, M. Bonnin, M. Chabot, S. Diaz-Tendero, F. Geslin, T. Hamelin, F. Hammache, C. Illescas, A. Jallat, A. Jorge, T. Launoy, T.K.C. Le, A. LePadellec, F. Martin, A. Meyer, L. Perrot, T. Pino, B. Pons, N. de Séreville, and K. Béroff. Excitation and fragmentation in high velocity C_nN^+ - He collisions. *Journal of Physics: Conference Series*, 875(11):102022, 2017
 6. Thibaut Launoy, Karine Béroff, Marin Chabot, Guillaume Martinet, Arnaud Le Padellec, Thomas Pino, Sandra Bouneau, Nathalie Vaeck, Jacques Liévin, Géraldine Féraud, Jérôme Loreau, and Thejus Mahajan. Ion-pair dissociation of

highly excited carbon clusters: Size and charge effects. *Phys. Rev. A*, 95:022711, Feb 2017

EMPLOYMENT

- **2020-present** Physics subject expert, Tutorwaves solutions.
- **2019-2020** Freelance Physics trainer for competitive examinations and Chess coach.

EDUCATION

- **Ph.D** Astrochemistry, September 2018
Université Paris-Sud(Université Paris-Saclay)Orsay campus, France.
Thesis Topic : Excitation and fragmentation of C_nN^+ ($n=1-3$) molecules in collision with He atoms at intermediate velocity; fundamental aspects and application to astrochemistry.
Advisor : Dr. Karine Béroff; Directrice de recherche, CNRS; Molecular systems, Astrophysics and Environment; ISMO
- **M.Sc Physics**, May 2014
National Institute of Technology, Calicut, India.
Thesis Topic: Numerical Solutions to Free and Confined Simple Atomic Systems Using Numerov's Method.
Advisor: Dr. Hari Varma, Associate Professor, IIT Mandi.
- **B.Sc. Physics**, May 2012
University of Calicut, India.
Thesis Topic : Introduction of Center of Charge in Electrostatics.
Advisor : Dr. M Raveendran, Professor (retired), University of Calicut, India.

COMPUTER SKILLS

Fortran90 and 77, C++ , Linux esp. Bash scripting, Linux cluster computing. , xmgrace scripting and plotting, LaTeX, Python (1 semester course in B.Sc.), Mat-Lab (1 semester course in M.Sc.), IRAF and VIREO (Astronomical simulation packages)

LANGUAGES

English : Excellent proficiency in speaking, reading and writing
French : Workable proficiency in comprehension and reading.
German : Beginner level. Actively learning.
Malayalam: Mother tongue.

CONFERENCES ATTENDED

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| 2017 | Astrochemical conference KIDA2017
Presented poster, Data of fragmentation for neutral, cationic and anionic C_nN molecules ($n=1-3$).
Bordeaux, France |
| 2016 | IMAMPC 2016 : International Meeting on Atomic and Molecular Physics and Chemistry
Presented poster, Excitation and fragmentation of C_nN molecules; fundamental aspects and application to astrochemistry
Le Havre, France. |
| 2016 | EMIE 2016 : Conference of researchers in France for systems of biological, relevant aggregates, molecules and complexes in atmospheric sciences and astrophysical interest
Dinard, France. |
| 2012 | Winter School in astronomy and Astrophysics 2012, Indian Institute of Space Science and Technology, Trivandrum, India |

HONORS AND AWARDS	2018	Ph.D Astrochemistry
	2015	3 year funding for Ph.D from École Doctorale Ondes et Matière (EDOM), Université Paris-Saclay
	2014	Qualified GATE-2014 (Graduate Aptitude Test for Engineering), All India Rank = 275
	2009–2014	INSPIRE-SHE Scholarship Department of Science and Technology(DST), Govt. of India
	2009	Central Sector Scholarship, Govt. of India

REFERENCES

- **Dr. Karine Béroff**
Directrice de recherche, CNRS; Molecular systems, Astrophysics and Environment; ISMO email: karine.beroff@u-psud.fr
- **Dr. Bernard Pons**
Professeur, Centre Lasers Intenses et Applications, Université de Bordeaux
email: bernard.pons@u-bordeaux.fr
- **Dr. Clara Illescas**
Profesora Titular, Facultad de Ciencias, Universidad Autónoma Madrid, Madrid
email: clara.illescas@uam.es

RÉPUBLIQUE FRANÇAISE

MINISTÈRE DE L'ÉDUCATION NATIONALE, DE L'ENSEIGNEMENT SUPÉRIEUR ET DE LA RECHERCHE

université
PARIS-SACLAY

DOCTORAT

Vu le code de l'éducation, notamment ses articles L.612-7, L. 613-1, D. 613-3 et D. 613-6 ;

Vu le code de la recherche, notamment son article L.412-1 ;

Vu l'arrêté du 25 mai 2016 fixant le cadre national de la formation et les modalités conduisant à la délivrance du diplôme national de doctorat ;

Vu l'arrêté du 10 juillet 2015 portant accréditation de l'école doctorale ;

Vu les pièces justificatives produites par Monsieur Thejus MAHAJAN, né le 30 septembre 1992, à Thirissur (Inde) en vue de son inscription en doctorat ;

Vu la procès-verbal du jury attestant que l'intéressé a soutenu, le 28 septembre 2018, une thèse portant sur le sujet suivant : **Excitation et fragmentation des molécules C₆H⁺ (n = 1-3) en collision avec des atomes de He à vitesse intermédiaire: aspects fondamentaux et application à l'astrochimie préparée au sein de l'école doctorale Ondes et Matière** devant un jury présidé par M. Alain DUBOIS et composé de Mme Karine BEROFF, M. Patrick ROUSSEAU, M. Jean Christophe LOISON, Mme Séverine BOYÉ-PÉRONNE ;

Vu la délibération du jury :

Le diplôme de **DOCTORAT** en **lasers, molécules, rayonnement atmosphérique préparé à l'université Paris-Sud** est délivré à Monsieur **Thejus MAHAJAN** au titre de l'année universitaire 2018-2019 et confère le **grade de docteur**, pour en jouir avec les droits et prérogatives qui y sont attachés.

Fait le 05 février 2019,

Le titulaire

La Présidente
de l'université Paris-Saclay

La Rectrice d'Académie,
Chancellerie des universités

SYME RETAILLEAU

Chantal AVENEL

Numéro du diplôme : 2018SACLS3112019036000005446

राष्ट्रीय प्रौद्योगिकी संस्थान कालिकट
National Institute of Technology Calicut



Upon the recommendation of the Senate hereby confers on

THEJUS MAHAJAN

the degree of

Master of Science

in Physics

in the year 2014

and places him/her in First Class with

Distinction

Given at the National Institute of Technology Calicut

on 10th January, 2015


Registrar


Director

യൂണിവേഴ്സിറ്റി ഓഫ് കാലിക്കറ്റ്



UNIVERSITY OF CALICUT
FACULTY OF SCIENCE

Whereas it has been certified by the duly appointed Examiners that

Thejus Mahajan

is qualified to receive the Degree of Bachelor of Science (B.Sc.) in

Physics

he having successfully completed the programme in April 2012

with Register No. SKAJSPH033

and been placed in overall grade B+

with Complementary Courses *Mathematics* and *Chemistry*

The Senate of the University of Calicut hereby confers on him

the Degree of

Bachelor of Science

with all the Rights, Privileges and Honours thereunto appertaining.

Given under the seal of the University

Cellusana

Vice Chancellor

CALICUT UNIVERSITY P.O.

Kerala, INDIA

PIN - 673 635

Date..22/02/2014.....

Digitally signed by DR. M ABDUL SALAM

Date: 22/02/2014 04:36:30 PM



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